

DFIR COPILOT

Digital Forensic Investigation Report

Case Number	DFIR-2026-127158	Status	Open
Case Name	Operation Paper Trail (Safe Demo)	Investigator	Vasanth Kumar
Created	2026-08-10T14:29:28+00:00	Generated	2026-08-10T14:29:28+00:00

Scope: Synthetic forensic exercise: suspicious download, PowerShell execution, outbound communication, persistence, and deleted-file evidence. No live malware or routable malicious infrastructure is included.

Executive Findings

Critical — Suspicious download-to-execution chain

A downloaded file was followed by PowerShell execution and an outbound connection within a short time window. This sequence is consistent with staged payload execution.

References: ART-48C3BDA5, ART-0D073843, ART-17AC3554 | MITRE: T1105, T1059.001, T1071.001 | Confidence: 95%

High — Potential anti-forensic file deletion

A deleted-file artifact was recorded after suspicious outbound activity.

References: ART-17AC3554, ART-5D588FAE | MITRE: T1070.004 | Confidence: 84%

High — High-risk artifact: Security Detection

Defender recorded suspicious PowerShell behavior in the synthetic exercise

References: ART-7DC75920 | MITRE: T1059.001 | Confidence: 80%

High — High-risk artifact: Service Installation

TrainingUpdate service was registered from a user-writable directory

References: ART-7718E2F8 | MITRE: T1543.003 | Confidence: 80%

High — High-risk artifact: YARA-X Match

PowerShell download behavior detected

References: ART-ABF43613 | MITRE: T1105 | Confidence: 80%

High — High-risk artifact: YARA-X Match

PowerShell download behavior detected

References: ART-7CC9350D | MITRE: T1105 | Confidence: 80%

Evidence Inventory

ID	Filename	SHA-256	Size	Verified
E-001	windows_events.json	487510716e0926381ef2971471ba36b01cb134bcc44d8cdd3bec6c04976f1aae	2969	Yes
E-002	autopsy_deleted_files.csv	e18bc113e63a24e63154a1c5d9385d934aca9d19141f2c8a11a8d902b5b950ec	319	Yes
E-003	invoice_viewer.ps1.txt	ea1f18898d65a3f176fc7e75e803187bb222df7e1d3f4dbea37f5f61ce6e66b1	407	Yes

Investigation Timeline

Time (UTC)	Severity	Artifact	Description	Ref
2026-08-09T10:15:02+00:00	Medium	Browser Download	User downloaded invoice_viewer.zip from hxxps://updates-example.invalid/invoice_viewer.zip	ART-48C3BDA5
2026-08-09T10:16:11+00:00	Low	File Creation	invoice_viewer.ps1.txt was created in the Downloads folder after archive extraction	ART-A53D98AC
2026-08-09T10:17:08+00:00	High	PowerShell Execution	PowerShell executed a training script containing Invoke-WebRequest	ART-0D073843
2026-08-09T10:17:31+00:00	Medium	Process Creation	A child process was created by powershell.exe	ART-61F9AF2C
2026-08-09T10:18:04+00:00	High	Network Connection	PowerShell established an outbound HTTPS connection to documentation IP 203.0.113.77	ART-17AC3554
2026-08-09T10:18:15+00:00	Medium	DNS Query	DNS query for updates-example.invalid	ART-D0C9E2AF
2026-08-09T10:20:42+00:00	High	Security Detection	Defender recorded suspicious PowerShell behavior in the synthetic exercise	ART-7DC75920
2026-08-09T10:22:19+00:00	High	Service Installation	TrainingUpdate service was registered from a user-writable directory	ART-7718E2F8
2026-08-09T10:25:36+00:00	Medium	Deleted File	Deleted-file record identified: invoice_viewer.ps1.txt	ART-5D588FAE
2026-08-09T10:26:01+00:00	Medium	Deleted File	Deleted-file record identified: payload.bin	ART-ACC0B199
2026-08-10T13:34:42+00:00	Info	File Metadata	Metadata recorded for E-001_windows_events.json (2,969 bytes)	ART-4E6960E8
2026-08-10T13:34:42+00:00	High	YARA-X Match	PowerShell download behavior detected	ART-ABF43613
2026-08-10T13:34:42+00:00	Info	File Metadata	Metadata recorded for E-002_autopsy_deleted_files.csv (319 bytes)	ART-B6CEF452
2026-08-10T13:34:42+00:00	Info	File Metadata	Metadata recorded for E-003_invoice_viewer.ps1.txt (407 bytes)	ART-DC421C10
2026-08-10T13:34:42+00:00	High	YARA-X Match	PowerShell download behavior detected	ART-7CC9350D

Chain of Custody

Time	Evidence	Action	Actor	Notes
2026-08-10T14:29:28+00:00	E-001	Imported and hash verified	Vasanth Kumar	SHA-256: 487510716e0926381ef2971471ba36b01cb134bcc44d8cdd3bec6c04976f1aae
2026-08-10T14:29:28+00:00	E-002	Imported and hash verified	Vasanth Kumar	SHA-256: e18bc113e63a24e63154a1c5d9385d934aca9d19141f2c8a11a8d902b5b950ec
2026-08-10T14:29:28+00:00	E-003	Imported and hash verified	Vasanth Kumar	SHA-256: ea1f18898d65a3f176fc7e75e803187bb222df7e1d3f4dbea37f5f61ce6e66b1
2026-08-10T14:29:28+00:00	E-001	Integrity verification	DFIR Copilot Demo Engine	Hash matched
2026-08-10T14:29:28+00:00	E-002	Integrity verification	DFIR Copilot Demo Engine	Hash matched
2026-08-10T14:29:28+00:00	E-003	Integrity verification	DFIR Copilot Demo Engine	Hash matched

Analyst note: AI-generated assistance must be independently validated against the cited evidence. This report records deterministic artifacts and correlations; it does not replace examiner judgment.

Examiner sign-off: _____ Date: _____